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Interpreting past epistemic modals in English, Dutch, and French

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According to many authors (Cinque 1999; Hacquard 2006, a.o.), epistemic modals in sentences such as “It had to be raining last night” always scope over past tense, expressing a present epistemic claim about a past event. However, there is no consensus on this point: for others (e.g. Rullmann & Matthewson 2018), epistemics can, or even must, scope below tense. This debate has substantial consequences for theories of modals. As of now, research lacks a systematic empirical description of the crosslinguistic picture. The examples discussed in the literature vary widely between languages, and most of the data comes from researchers’ own intuitions, fieldwork involving a limited number of informants, or informal questionnaires. The goal of this paper is, first, to settle the current disagreement on the judgments, by comparing in a more controlled way judgments in three languages, English, Dutch, and French. Second, it is to assess the prediction from van Dooren (2020b) that there is variation between languages, which can be explained by crosslinguistic differences in modals’ lexical status: (semi-)auxiliaries versus verbs. We report two acceptability judgment experiments testing past tensed epistemic modal sentences’ interpretations in English, Dutch, and French, comparing *had-to*/*moest*/*devait* to the verbs *seemed*/*leek*/*semblait*, using identical sentences and methods and controlling for various factors overlooked in previous studies. We show that (i) epi>tense is available in all languages; (ii) English *had-to* is overall dispreferred as compared to Dutch *moest* and French *devait*. While we only find nuanced support for van Dooren’s proposal, our results open promising avenues for more controlled experimental research on modal-temporal interaction.



1 Introduction

When epistemic modals bear past tense morphology as in (1a-c), two different scope interpretations are logically possible depending on the relative order of the modal and past tense: either a current epistemic claim about a past event (“given the evidence now, Al was in the bathroom”), if the modal scopes over tense (epi > tense, (1)), or a past epistemic claim (“given the evidence available before speech time, Al was in the bathroom”), if the modal scopes under tense (tense > epi, (1)).

- (1)
- | | | | | | |
|-----|---|---------------|---------------|------------------|----------------|
| a. | Al | had to | be | in the bathroom. | <i>English</i> |
| b. | Al | moest | op de wc | zijn. | <i>Dutch</i> |
| | Al | must.PST | on the toilet | be.INF | |
| c. | Al | devait | être | aux toilettes. | <i>French</i> |
| | Al | must.PST | be. INF | on-the toilet | |
| i. | ‘Given the evidence now, it is necessary that Al was in the bathroom.’ epi > tense | | | | |
| ii. | ‘Given the evidence in the past, it was necessary that Al was in the bathroom at that time.’ tense > epi | | | | |

Do epistemic modals scope above or below tense in sentences like (1)? This question is debated. According to many researchers, sentences like (1) admit only the epi > tense interpretation, (1-i), where epistemic modals scope over tense (Groenendijk & Stokhof 1975; Iatridou 1990; Condoravdi 2002; Stowell 2004; Hacquard 2006; 2010; Ter Beek 2008; Laca 2012, a.o.). However, other authors—drawing on evidence from a wider range of sources and languages—argue that epistemic modals can, or even must, take scope under tense (von Stechow & Gillies 2007; Boogaart 2007; Martin 2011; Aelbrecht 2010; Homer 2013; Rullmann & Matthewson 2018, a.o.).

Distinguishing between readings (1-i) and (1-ii) is not straightforward: Most naturally occurring contexts do not clearly distinguish between the two interpretations, since what was taken to be true in the past is most often still believed to be true in the present—if I now think that Al was in the bathroom, so did I most of the time think before. But this is not an innocuous controversy. These data play a key role in a fundamental theoretical puzzle: what is the proper syntactic analysis for modals?

Sentences like (1) have been used to build and argue for or against various syntactic theories of modals. One of the puzzles researchers focus on is what underlies the apparent differences between epistemic modality—expressing possibility/necessity based on evidence or knowledge, as in (1a-c)—and other types of modality. In English and many other languages, modals convey various ‘flavors’ of modalities. A common distinction is between epistemic modality and ‘root’ modality (a term introduced by Hofmann 1976), the latter covering all non-epistemic types like deontic modality, which expresses necessity based on rules (see (2)). One remarkable fact is that when modals like English *had-to* get root interpretations, they have to be interpreted in the scope of tense: sentences (2a-c) express *past* obligations (“given the rules at the time, it *was* necessary

that Al was home by 10”), rather than *present* ones (“given the rules now, it is necessary that Al was home by 10”).

- | | | | | | |
|-----|-----|--|---------------|---|------------------|
| (2) | a. | Al | had to | be home by 10. (His parents told him so). | <i>English</i> |
| | b. | Al | moest | om 10 uur thuis zijn. | <i>Dutch</i> |
| | | Al | must.PST | at 10 hour home be.INF | |
| | c. | Al | devait | être rentré à 10 heures. | <i>French</i> |
| | | Al | must.PST | be.INF home at 10 hours | |
| | i. | #‘Given the current rules, Al is required to have been home by 10.’ | | | #deontic > tense |
| | ii. | ‘Given the past rules, Al was required to be home by 10.’ | | | tense > deontic |

The distinction between epistemics and roots is supported by a range of diagnostics, both at the syntactic and the semantic level. This includes their scope interaction with subjects (e.g., Ross 1969; Brennan 1993), patterns of so-called ‘modal stacking’ (e.g. Picallo 1990), and their interaction with negation (e.g., Cormack & Smith 2002) (see e.g. Hacquard 2011; Barbiers & van Dooren 2017, for reviews). These scope differences are captured, notably, in Cinque’s (1999) hierarchy, which stipulates a universal ordering of functional elements where (among others) epistemic modals always scope higher than Tense and Aspect, and roots scope below. But empirically, is it really the case that epistemic modals universally scope over tense, and roots below? When we look at more diverse languages or investigate naturalistic uses, the answer is not so clear. If we find epistemics scoping under tense, this requires us to reconsider Cinque’s generalization.

The goal of this paper is, first, to settle the disagreement on the judgments for sentences such as (1a-c), by comparing in a systematic and controlled way English, Dutch, and French, and to investigate how stable they are across languages and speakers. Currently, research lacks a systematic empirical description of the crosslinguistic picture: the examples discussed in the literature vary widely between languages, making it difficult to compare between them. Past epistemic modal sentences are not so frequent in spontaneous productions, and the problem is exacerbated by the subtle nature of these judgments. Our second, more specific goal, is to assess a prediction from van Dooren (2020b) that there is variation between languages, which is due to a difference in the lexical status of modals in Dutch, English and French: (semi-)auxiliaries (functional elements) vs verbs (lexical elements).

We conducted two acceptability judgment experiments comparing how sentences with past epistemic modals like (1) are interpreted by native speakers of English, Dutch, and French. We used identical materials and methods in the three languages, controlling for various factors sometimes overlooked in previous studies, in particular the competition with a non-epistemic interpretation, and the presence/absence of narrative context. In the first experiment (‘offline’

judgments), participants had unlimited time to make their judgments. In the second experiment ('online' judgments), run as a follow-up, participants were instructed to answer as quickly as possible, an attempt to prevent their potential reinterpretation strategies and get at their primary interpretation (e.g. Bott & Noveck 2004; see Bader & Häussler 2010, a.o.).

Results from Experiment 1 suggest that the *epi* > tense interpretation, (1-i), is available in all three languages, supporting the view that crosslinguistically, epistemic modals can be interpreted above the tense of their complement (Groenendijk & Stokhof 1975; Iatridou 1990; Stowell 2004; Condoravdi 2002; Hacquard 2006; Ter Beek 2008; Laca 2012 a.o.). Yet, our results reveal differences between languages. First, the pattern we find for Dutch suggests that it also allows the second interpretation, (1), suggesting that in Dutch, epistemics can both scope below and under past. This is in line with van Dooren (2020b), who predicts that (1) is available in Dutch but not English and French. Second, the comparison between languages reveals that—when tested in epistemic scenarios—English *had-to* is overall less acceptable than its Dutch and French counterparts *moest* and *devait*. This finding is highly relevant, since it might explain part of the disagreement: one of Rullmann & Matthewson's (2018: 326) key arguments to support their view is that *had-to* examples sound unnatural to English native speakers when presented in a context that favors present epistemic claims. Instead, we propose that the lower acceptability we find for *had-to* does not result from scope considerations, but rather from the specific licensing conditions for epistemic *have-to* (Williams et al., in prog.).

Results from Experiment 2 are harder to interpret. We only partially replicate Experiment 1's results: the main first result from Experiment 1, as well as the lower acceptability for *had-to*, but it does not confirm the difference between Dutch and the other languages. Various factors might have contributed to this result, as we discuss at the end. This opens interesting avenues for future experimental research on modal temporal interaction.

Before we proceed, note that our study's empirical scope is limited. In particular, the literature on modal-temporal interaction often discusses two notions together: *Temporal Perspective* (TP) and *Temporal Orientation* (TO), following Condoravdi (2002). The *Temporal Perspective* (TP)—which we focus on in this paper—refers to the time of evaluation of a modal (technically, the time when its conversational background is assessed). It can be past (e.g., "Al had to come"), present (e.g., "Al has to come"), or future (e.g., "Al will have to come"). One way to recast our question is to say that we investigate a proposed restriction on TP: the generalization that epistemics always have a present TP, whereas root modals' TP is set by tense (Groenendijk & Stokhof 1975; Stowell 2004; Abusch 1997; Hacquard 2006, a.o.). The *Temporal Orientation* (TO) refers to the relation between the time of evaluation of the modal and the time of the eventuality described by its complement, often determined by its aspectual properties; the TO can be past-oriented (e.g., "Al must have been home"), present-oriented (e.g., "Al must be home"), or future-oriented ("Al must

go home”).¹ In this paper, we put aside debates about TO and focus on the generalization about TP. Relatedly, we do not discuss thoroughly grammatical aspect (which largely determines TO) and examples where perfect aspect appears under the auxiliary (e.g., “Al must have been sick”)—though (at least in English), the latter may be a more natural way to convey present epistemic claims.

The rest of the paper is organized as follows. In **Section 2**, we review key datapoints from the literature, first in English, then in Dutch, and finally in French. We show how the current lack of consensus about the judgments justifies a controlled experiment. In **Section 3**, we present three different syntactic theories and the predictions they make about the possible available interpretations of sentences like (1) (Hacquard 2006; 2010; Rullmann & Matthewson 2018 and van Dooren 2020b). In **Section 4**, we present Experiment 1, where participants had unlimited time to make their judgments, and in **Section 5**, Experiment 2, in which participants were instructed to answer as quickly as possible. **Section 6** discusses the consequences of our results for syntactic theories of modals crosslinguistically and open questions for future research.

2 The data

Competing claims have been made for English, Dutch, and French. In the section below, we present and discuss the datapoints that have been discussed in the literature, per language.

2.1 English *had-to*

According to many authors, English epistemic modals scope over tense (Groenendijk & Stokhof 1975:68—69; Iatridou 1990; Stowell 2004; Hacquard 2006; 2010; 2011; Hacquard & Cournane 2016, a.o.). Stowell (2004) originally discussed the example in (3):²

- (3) There **had to** be at least a hundred people there. (Stowell 2004:626)

¹ In terms of TO, it has been argued that root modals are restricted to future TO, whereas epistemics can have present and past TO (Condoravdi 2002; Rullmann & Matthewson 2018; Werner 2006; Klecha 2016). Condoravdi (2002) originally discussed the readings of English *might have* in sentences like (i)/(ii). On reading (i), *might have* expresses present epistemic uncertainty about a past event. On reading (ii), it expresses that at some past time it was metaphysically possible for the world to develop in such a way that the event would happen. Crucially, these represent only a subset of the logically possible combinations of TP and TO. To focus on TP, in this paper, we leave aside discussion of these examples.

(i) John might have won the game (but I’m not sure if he did).

(ii) John might have won the game (if he hadn’t been feeling sick that day).

² Recall that the reason why English examples involve the semi-auxiliary *have-to* (rather than *must*) here is that modal auxiliaries like *must* do not inflect for tense (**musted*). Accordingly, a more natural way to convey present epistemic claims is to use perfect aspect under the modal (e.g., “There must have been at least a hundred people there”).

Stowell (2004) claims that sentences like (3) express a present epistemic claim: ‘given the evidence available now, it is likely that (in the past) there were at least a hundred people there’. Stowell contrasts this interpretation with root uses of modals—as we saw in (2), with a root interpretation, *had-to* is uncontroversially interpreted under tense.

Example (3) in isolation does not prove that the tense > epi interpretation is not available. Indeed, the sentence is both acceptable whether the modal scopes over or under tense, since a past epistemic claim (‘given the evidence available in the past, it was likely that there were a hundred people there’) could still hold in the present. To tease apart the two readings, we need to consider some more elaborate scenarios, like (4) (introduced by Hacquard & Cournane 2016). In (4), the speaker—a detective investigating a crime—used to believe some fact P (e.g., P: “Al was in Montreal at the time of the crime”), but then receives some new evidence that P was false and that an alternative fact, Q (incompatible with P), is in fact true (e.g., Q: “Al was in DC”). Crucially, in such scenarios, past and present epistemic claims (paraphrasable by, respectively, “I used to believe P” vs “I now know P”), cannot both be true at the same time. Hacquard & Cournane claim that in such contexts, “it had to Q” is acceptable. They contrast *had-to* with *seemed*, which (uncontroversially) scopes below tense and shows the opposite pattern: acceptable in (4), but not in (4), where one has to use present tense (“Al seems to have been in DC last night”).³

- (4) [Context: Al has been a prime suspect for a crime that occurred last night in Montreal. Up to now, all of the evidence pointed to him being in Montreal last night. But just now, the detective receives fresh evidence that proves that Al was in fact in DC last night.]
- a. Al {**had to**/**#seemed to**} be in DC last night. Present epistemic claim (epi > tense)
 - b. Al {**#had to**/**seemed to**} be in Montreal last night. Past epistemic claim (tense > epi)
- (Hacquard & Cournane 2016)

As put forward in the introduction, the view according to which epistemic modals scope over tense is disputed. Rather, a number of authors argue that epistemic modals can (von Stechow & Gillies 2007), or must (Rullmann & Matthewson 2018) scope under tense. Their claim is based on several arguments.

The first point, made by Rullmann & Matthewson (2018), is that the examples used as evidence that *had-to* expresses present epistemic claims, like (3)/(4a), sound “(at best) marginal” to native speakers (Rullmann & Matthewson 2018: 326). To support this, they ran an informal questionnaire asking native English speakers to evaluate *had-to* sentences in contexts that force the

³ Hacquard & Cournane (2016) only argue that epistemic modals can outscope tense, not necessarily that they must, so they don’t make the claim that “had to P” is false in (4b), only that “had to Q” is true in (4a).

epi > tense interpretations, like (5). They report that 3/8 speakers judge (5) marginal and only 3/8 accept Stowell's example (3) presented in a context that supports a present epistemic claim (6).⁴

- (5) [Context: Up until just now, all of the evidence pointed to Mary being home last night. But now, fresh evidence proves that Mary's home was empty last night.]
Mary **had to be out** last night. (Rullmann & Matthewson 2018:146)
- (6) [Context: You are telling someone about how many people were at a party that was held last night. You were at the party and while it was going on, you thought there were only about fifty people there. But now it's the next day and you are cleaning up, and based on the number of dirty glasses and other evidence, you realize it must have been more like 100 people. You say:]
There **had to be** a hundred people here. (Rullmann & Matthewson 2018:147)

Second, Rullmann & Matthewson (2018) provide naturalistic corpus examples (from the COCA Corpus, Davies 2008–2015), which, they argue, can only be taken to convey past epistemic claims and not present ones. As support, they ran another informal questionnaire (n = 11), showing most speakers accept *had-to* sentences in contexts like (7).

- (7) This morning I opened my phone bill and was shocked when I saw that I owed \$10,000. This **had to be** a mistake! Unfortunately, it turned out to be correct [...].
(Rullmann & Matthewson 2018:61)

Many authors (Boogaart 2007; Hacquard 2006; 2011; Laca 2012 for French) propose to treat such cases as instances of Free Indirect Discourse (FID), a mode of narration where we “listen to a protagonist's thought” (Eckardt 2014). As we will see in Section 2.3 and 2.4, very similar examples (i.e., cases where, within narrative context, past epistemic modals seem to be interpreted as past epistemic claims and cannot be interpreted as present ones) have been discussed for Dutch and French.

2.2 Free Indirect Discourse and epistemic modals

FID is a mode of narration where the perspective is shifted to one of the agents in a story, without it being overtly marked (Banfield 1982; Schlenker 2004; Eckardt 2015; Maier 2015; Charnavel 2025; see review in Reboul et al. 2016). There are two main types of approaches to FID, one in which it more closely resembles direct discourse (Banfield 1982; Schlenker 2004; Eckardt 2015; Maier 2015), and one in which it more closely resembles *indirect discourse* (Doron 1991, Sharvit

⁴ Rullmann & Matthewson don't report details about the procedure (neither the full list of sentences, nor the phrasing of the instructions), making it difficult to discuss the data further.

2008; Charnavel 2025).⁵ Of course, here we by no means intend to do an exhaustive overview of the literature on FID. We focus on the parts that are crucial for the puzzle at hand: Do examples like (7) show that epistemic modals scope under tense?

First, there is no real disagreement regarding the naturalness proper of examples like (7). They are generally judged acceptable, by linguists, their informants, as well as naïve native speakers (at least when presented in narrative/story-telling contexts, as Rullmann & Matthewson’s (2018) informal study shows). Furthermore, they are attested in corpus data, as evidenced by Rullmann & Matthewson (2018) and Reisinger et al. (2022). In particular, Reisinger et al. (2022), in a larger corpus-based study on English, provide multiple examples of FID involving *might have*, *may have*, and *must have*—which they categorize as having “past TP”.⁶

The first point of disagreement is however how to describe such cases. In FID contexts, there are two temporal reference points to distinguish: the speaker’s (uttering the sentence *now*), and the protagonist’s (the character in the story). Depending on whether one defines the TP as relative to the speaker’s, or to the protagonist’s reference point, there are two ways to characterize TP: if we define the TP as relative to the speaker, we should say that the TP of *had-to* is past—as the epistemic claim is made before speech time (e.g. Rullmann & Matthewson 2018; Reisinger et al. 2022). Others (e.g. Hacquard 2011; Falaus & Laca 2020), say that in those examples, the TP is present—because it is *simultaneous* to the time of the *protagonist*’s context.

The second, but deeper, point of disagreement, is whether examples like (7) can be taken to falsify the claim that epistemic modals only scope above tense. For Hacquard (2006; 2011), it doesn’t, because the past tense morpheme does not have a semantic backshifting effect. Crucially, the epistemic claim is made simultaneous to the time of the context, which is set in the past. Hacquard proposes a similar mechanism as when epistemic modals appear embedded under past attitude verbs, as in (8): since the modal is under the attitude verb, its local time of evaluation is shifted to the past, and we obtain an epistemic claim at the time of the thinking. For her, in both (7) and (8), the past tense morpheme does not carry semantic tense as the epistemic claim is not further backshifted from the tense in the main clause; it is instead the result of the sequence of tense effect (Abusch 1997; Ogihara 2011).

(8) Two days ago, Poirot thought that Mary **had to** be the murderer. (Hacquard 2011:38)

⁵ FID can be described a mixture of direct and indirect speech: exclamatives and interrogatives are allowed, and one uses the protagonists’s vocabulary, similarly to direct speech. Unlike direct speech however, and closer to indirect speech, tense and pronouns are interpreted relative to the protagonist, and not the speaker.

⁶ In their study, Reisinger et al. (2022) focus on six modal forms: *may*, *must*, *might*, *may-have*, *must-have*, and *might-have*. For each form, they analyze 100 tokens, annotating them for flavor, embedding status, tense of the matrix verb, temporal parameters (TP/TO), and narrative perspective (narrator/protagonist/ambiguous). They provide several examples involving *might-have*, *may-have*, and *must-have*. Note however that their study does not include *had-to*.

A challenge for Hacquard’s proposal is that, as shown by Banfield (1982), it might not be possible to reduce FID to sentence embedding for independent reasons.⁷ But this is an ongoing debate. For instance, Charnavel (2025) presents an analysis of FID where, instead of a verb, there is a logophoric operator that semantically embeds FID (Charnavel 2019; 2020), binding the indexicals. Charnavel claims that past tense in FID in a language should be analyzed as a zero tense if sequence of tense effects exists in that language, making explicit the parallel between FID and indirect discourse. Falaus & Laca (2020), similarly, argue for a semantically zero tense in cases of epistemic modals in FID and embedded contexts. Unlike Hacquard however, they argue that past epistemic modals in FID falsify the claim that epistemic modals cannot scope under tense: A zero tense, similar to a regular tense, should be hosted in a TP phrase.

Given the lack of consensus on how to generally analyze FID, it is not easy to decide what to conclude from examples like (7). Do they constitute evidence that epistemic modals scope under tense? For the remainder of this data section, we will put aside such examples involving overt narrative contexts, and focus on the question whether, beyond those cases, we can find convincing examples of past epistemic modals with past TP.⁸

2.3 Dutch *moest*

In Dutch, it has also both been argued that epistemic modals scope over (Ter Beek 2008) and under tense (Boogaart 2007; Aelbrecht 2010; Rullmann & Matthewson 2018). But the data exhibit interesting differences.

According to Boogaart (2007), the possible interpretations of Dutch epistemic modals constitute the “mirror image of English” (Boogaart 2007:51). For him, sentences where *moeten* ‘must’ bears past tense, like (9), can express past epistemic claims, but not current ones. Rullmann & Matthewson (2018) also provide examples where *moest* has to be interpreted under tense, like (10) which asserts that at some time (yesterday) preceding the utterance time (now), it was epistemically necessary for the speaker that the key was in the drawer.

- (9) Dutch
Hij **moest** ziek zijn. (Boogaart 2007: 50)
He must.PST ill be
i. #‘it is very likely that he was ill.’ epi > tense
ii. ‘it was very likely that he was ill.’⁹ tense > epi

⁷ We thank a reviewer for pointing this out. Banfield (1982) shows in particular that FID does not exhibit the syntactic complementation properties of indirect discourse—notably, in FID, we observe inverted questions, exclamations, repetitions or incomplete clauses, which are characteristic of root clauses.

⁸ While we put the analysis of FID-examples outside of the scope of this paper, note that this is why in our experiment (to be presented in Section 4), we try to control for FID by presenting critical sentences as direct speech.

⁹ Here and in (12)/(13), we use the paraphrase originally used by the author (instead of “given the past/present evidence” as in (1i–ii)). The exact paraphrase makes no difference for our purposes; in terms of scope, they are equivalent.

- (10) [Context: Yesterday, when I wanted to go to work, I couldn't find my key anywhere. I tried to remember where I might have left it the previous night. I felt in the pocket of my pants, looked in my nightstand, even searched the waste basket, but all in vain. Suddenly I knew.]

De sleutel **moest** (wel)¹⁰ in de la liggen.

the key must.PST (DISC) in the drawer lie.INF

'The key **had to** be in the drawer.' (Rullmann & Matthewson 2018:25)

Both cases could be instances of FID, similar to (7). Yet, Aelbrecht (2010) provides another example, (11), that suggests that even outside past narrative discourse contexts, *moest* can be interpreted under tense. In (11), the question under discussion is whether a man, on his birthday, was in Portugal. The second part of the sentence ("but the new evidence...") establishes that we now know he was not: therefore, the claim 'he was in Portugal' can't be made at the present. Still, *moest* is acceptable. Crucially, contrary to (10), there is no context that shifts (11) to the past. Aelbrecht argues that this shows that *moest* can scope over tense (it is interpreted in the scope of *yesterday*).¹¹

- (11) Gisteren **moest** hij nog in Portugal geweest zijn op zijn verjaardag, maar
yesterday must.PST he still in Portugal be.PST.PTCP be on his birthday but
het nieuwe bewijsmateriaal toont aan dat dat een foute conclusie was.
the new evidence shows on that that a wrong conclusion was
'Yesterday given the evidence, he was still in Portugal on his birthday, but the
new evidence indicates that that conclusion was wrong.' (Aelbrecht 2010:34, our
translation)

Other authors, notably Ter Beek (2008:108), claim that Dutch epistemic modals can embed tensed complements, i.e., they can scope over tense (she doesn't argue that they can't scope below). To show that epistemic modals can scope over tense, Ter Beek uses evidence from sentences in which the auxiliary *hebben* 'have' appears in the modal's complement, and from the interaction between modals and the future tense. Both arguments being more complex than the past modal sentences we focus on this paper, we do not review them here (but see van Dooren 2020b for discussion).

2.4 French *devait*

In French as well, it has both been argued that epistemic modals scope over tense (Hacquard 2006; 2011; Laca 2012) or under tense (Martin 2011; Homer 2013; Falaus & Laca 2020). Again, there is no clear consensus on the data.

¹⁰ The discourse particle *wel* helps obtaining an epistemic interpretation of *moest* (see Hogeweg 2009 for an overview of its uses).

¹¹ According to our informants, English *had-to* and French *devait* are infelicitous in similar sentences. This further suggests that example (11) is not analyzable as FID: as discussed in Section 2, in FID contexts, *had-to* and *devait* are generally acceptable.

In French, there are two main ways modals like *devoir* ‘must’ combine with past tense, illustrated in (12): the *imparfait* (*devait* in (12)), used for past imperfective, and the *passé composé* (*a dû* in (12)), which consists of an auxiliary and past participle, and is used either for past perfective or present perfect.¹²

- (12) a. Il **devait** pleuvoir. (Homer 2013:1)
 it must.PST.IPV rain.INF
- b. Il **a dû** pleuvoir (Homer 2013:2)
 it has must.PST.PTCP rain.INF
- i. ‘It is held certain (by me) now that it was raining then.’ epi > tense
 ii. ‘It was held certain (by me) then that it was raining then.’ tense > epi

Laca (2012) claims that both (12) are interpreted as present epistemic claims (i): here again, though appearing on the modal, past tense only scopes over the prejacent. By contrast, Homer (2013) argues that both interpretations are possible (see also Martin 2011, who rejects the view that epistemics scope over tense with the possibility modal *pouvoir*). As support, Homer provides example (13), which cannot be interpreted as a present epistemic claim, since the speaker cannot still be believing that the addressee had a heart attack.¹³ Homer further argues that (13) cannot be treated as FID because the sentence is used in the context of a conversation and directly targets the addressee with the indexical *you*. This argument seems flawed, however, because it is not true that FID is restricted to narrative past text (see Fludernik 1993, who argues that FID has spread into oral communication and is not restricted to narrative text); precisely in conversational contexts, indexical adverbials behave differently (Charnavel 2025).

- (13) [Context: On the day of the utterance D0, the speaker’s grandfather asks her why she panicked and stormed out of the house yelling on D-6, when she saw him lying on the floor. The man is 90 years old but the speaker knows at D0 that he has never had any health problem; right after her fit of panic on D-6, the speaker realized that her grandfather was in fact meditating on the floor.]
 Tu **devais** sûrement avoir eu une crise cardiaque.
 You must.PST surely have had a attack cardiac
 ‘It was held very likely/certain (by me) that you had had a heart attack.’ (Homer 2013:4)

¹² French *passé simple* (“Il **dut** pleuvoir”) is unambiguously used for perfective, but nowadays almost out of use, except in narrative contexts. We also leave aside *plus que parfait* (“Il **avait dû** pleuvoir” ‘It had to have rained’) and conditional morphology (“Il **aurait dû** pleuvoir” ‘it should have rained’); see Martin 2011; Homer 2013; Laca 2012 for discussion.

¹³ Homer’s (2013) second argument is based on scope paradoxes with negation, a topic we leave aside in this paper given the complexity of modals and negation interaction.

3 Is there a syntactic difference between epistemic and root modals?

3.1 Three theories of modal temporal interactions

What are the consequences for the syntactic analysis for modals? In this section, we focus on three (competing) syntactic analyses of modals which have used data from Section 2 as supporting evidence for their theories.¹⁴

As mentioned in the introduction, the literature on modals draws a major distinction between epistemic and ‘root’ (non-epistemic) flavors. The cross-linguistic generalization is that modals with epistemic interpretations tend to scope higher than root ones. This is notably captured in Cinque’s (1999) hierarchy in which epistemic modals appear in a higher functional projection than Tense, Aspect, and root modals (14).

(14) Modal epi > Tense > Aspect > Modal root (irrelevant projections omitted)

There is a tension here: on the one hand, given the cross-linguistically robust fact that in many languages, modals can be used for both flavors (Kratzer 1977), one may want to keep one single lexical entry for epistemics and root modals. But on the other hand, we need to capture the differences between epistemic and roots in how they interact with other scope bearing elements in the sentence, and the fact that not all scope interpretations are available. Our question, how modals interact with tense, is one piece of this general puzzle.

Kratzer (1981; 1991) originally proposed that root and epistemic modals share the same lexical entry, and that modal flavor is determined via two contextually determined conversational backgrounds, which provide the set of worlds the modal quantifies over. Yet, this doesn’t predict differences between roots and epistemics’ scope behaviors. Building on Kratzer, Hacquard (2006; 2010) offers a way to explain those differences while keeping one single entry for modals. For her, modal sentences are structurally ambiguous, with two possible underlying syntactic structures: one where the modal is high (above TP, (15)), the other where the modal is low (above VP, (15)). The main departure from Kratzer is in how the modal base is determined: modals have an event variable, which gets bound by the higher event of evaluation. Modals located above tense get an epistemic interpretation because the modal’s event variable is bound by the higher speech event, whereas modals located below tense get a root interpretation because the event variable is bound by the aspect quantifying over the VP event.

(15) a. high: Modal > Tense > Aspect > VP
b. low: Tense > Aspect > Modal > VP

¹⁴ We provide a schematic overview of the proposals discussed in this section at the end of the section, in **Table 1**. We obviously cannot do justice to all proposals (see Heine 1995; Nuyts 2001; Boogaart 2007; Martin 2011; Homer 2013, who contributed to this debate). We choose to focus on these three proposals because they directly respond to one another, and make opposing predictions.

Rullmann & Matthewson (2018) offer a potentially simpler analysis, aimed at capturing data from a wider range of languages (English, Dutch, Gitksan (Tsimshianic), and St’át’imcets (Salish)). They propose that both root and epistemic modals are in the same projection, below tense and above aspect (see also Homer 2013). They argue that all modals scope under tense and therefore receive past TP if the tense provides a past reference time. As we saw in 2.1, one of their empirical arguments is that epistemic *had-to*-sentences where tense is interpreted below the modal, like (3), are “marginal at best” in English, whereas speakers accept sentences where tense is interpreted above the modal, like (7) (FID contexts).

Van Dooren (2020b) proposes (contra Rullmann & Matthewson, following Hacquard) that epistemics scope above tense cross-linguistically, but that Hacquard’s account needs to be recast in terms of complement size (see also Ramchand 2014; Ramchand & Svenonius 2014). Her generalization is spelled out in (16) (from van Dooren 2020b):

- (16) Generalization: Modals combine with different sizes of complements depending on their modal flavor, namely:
- *epistemic* modals combine with a Tense Phrase (Cinque 1999; Hacquard 2006)
 - *root* modals combine with smaller phrases (a vP or smaller, depending on the sub-flavor of root).

Van Dooren proposes that the crucial difference between English and Dutch comes from the lexical status of the modal: while English modals are (semi-)auxiliaries (Jespersen 1909–1949), Dutch modals are verbs (as independently argued by Den Dikken & Hoekstra 1997; Barbiers 2006; Aelbrecht 2010; Broekhuis & Corver 2015; see Haeseryn et al. 1997; IJbema 2001; Van Riemsdijk 2002 for a contrasting view).¹⁵

¹⁵ The term semi-auxiliary is common in the literature, the idea being that “semi-auxiliaries” don’t behave like canonical auxiliaries (*must*, etc.), but also differ from lexical verbs (*want*, etc.). In English, where the label is used for *have-to/ought-to* (etc.), semi-auxiliaries differ from canonical auxiliaries in that they inflect for tense/agreement, require do-support in negation and yes/no questions. Unlike lexical verbs, they cannot assign argument structure (for instance, they can’t appear with direct objects) or stand alone outside of ellipsis contexts (they embed TPs or CPs). The label is also standard for French *devoir/pouvoir* (see footnote 16).

One argument van Dooren (2020b) uses to argue that Dutch modals are verbs is that they can introduce arguments (Barbiers 2006: 8). In (i), the PP is licensed by the modal and not the main verb under the modal, as (ii) shows: without the modal, the PP is not licensed. Note that in (i), the modal has a deontic interpretation. Examples with epistemic interpretations have not been attested.

- (i) Jan moet **van zijn vader** naar huis gaan.
 Jan must of his father to home go
 ‘His father forces Jan to go home.’

- (ii) Jan gaat (*van zijn vader) naar huis.
 Jan goes of his father to home
 ‘Jan is going home.’

(Barbiers 2006:8)

Modal (semi-)auxiliaries are functional elements inside a functional projection (FP, (17a)): they form a monoclausal structure with the main verb in the sentence. Modal verbs, on the other hand, head a verbal projection and project their own functional projections, including tense (17b).

- (17) a. [FP Modal auxiliary [FP [vP verb]]]
 b. [FP [vP Modal verb [FP [vP verb]]]]

The difference in lexical status, in combination with the generalization in (16), makes the following predictions. Modal (semi-)auxiliaries (like English *have-to*) allow only one scope interpretation, where the modal is interpreted above tense for epistemic flavors (18a), and where the modal is interpreted below tense for roots (18b). By contrast, modal verbs (like Dutch *moest*) head their own functional projections, including tense and aspect. Sentences with root modal verbs only have one tense phrase (because they select for at most a vP complement (16)), where the modal scopes below tense (19c). Epistemic modal verbs, however, combine with a TP-complement (19b); as a consequence, the sentence has two tense projections. Van Dooren therefore predicts two possible interpretations for Dutch epistemic modal sentences (contra English): (19a), with the modal scoping over tense, or (19b), with tense scoping above the modal.

- (18) Sam had to sleep.
 a. [TP Sam [FP have to_{epi} [TP -d [vP ~~Sam~~ sleep]]] epistemic > tense
 b. [TP Sam [TP -d [FP have to_{root} [vP Sam sleep]]] tense > root
- (19) Sam moest slapen.
 Sam must.PST sleep.INF
 a. [TP Sam [vP moet [TP -st [vP ~~Sam~~ slapen]]] epistemic > tense
 b. [TP Sam -st [vP moet [TP [vP ~~Sam~~ slapen]]] tense > epistemic
 c. [TP Sam -st [vP moet [vP ~~Sam~~ slapen]]] tense > root

Table 1 schematically summarizes these three proposals.

3.2 Cleaning the table

There is no general consensus on the judgments, and how they might vary across languages and speakers. The majority of the data comes from researchers' own intuitions, fieldwork involving a limited number of informants, or informal questionnaires, such as those employed by Rullmann & Matthewson (2018). Corpus data are not so informative: epistemic modals in the past are quite rare, with root modals being far more prevalent (see e.g., for English: van Dooren et al 2022; for Dutch: Nuyts 2001; van Dooren et al 2019; for French: Dieuleveut 2023), and amongst these rare cases, contexts that allow to tease apart the two readings in (1) (with new contradicting evidence) almost never occur (for discussion of how hard it is to determine subtle semantic distinctions

such as TP/TO or protagonist vs narrator’s perspective from corpus data, see Reisinger et al. 2022).

Authors	Generalization
Cinque 1999	Modal epistemic > Tense > Aspect > Modal root
Hacquard 2006; 2009	Modal sentences are structurally ambiguous: a. high: Modal > Tense > Aspect > VP b. low: Tense > Aspect > Modal > VP Modals located above tense get an epistemic interpretation because the modal’s event variable is bound by the higher speech event, whereas modals located below tense get a root interpretation because the event variable is bound by the aspect quantifying over the VP event.
Rullmann & Matthewson 2018	Tense > Modal > Aspect
van Dooren 2020b	Modal epistemic > Tense > Aspect > Modal root Root modals always scope under tense; epistemic modal auxiliaries always scope over tense. Epistemic modal verbs can scope either over or under tense.

Table 1: Simplified summary of the accounts.

There has been only one controlled experimental study on the topic. Van Dooren (2020a) used an online Truth-Value Judgment Task to assess judgments on past epistemic modal sentences in scenarios similar to Cournane & Hacquard’s (2016) example (4), comparing English (*had-to/seemed*) and Dutch (*moeten/leek*). In the experiment, participants were asked to indicate whether different sentences were true or false in a given context. The four conditions are spelled out in (20)a/b/c/d. In order to control for FID and favor epistemic (vs. root) interpretations, the sentences were presented as uttered by a detective, in direct discourse. Since *seem/leek*, as lexical verbs, (only) scope under tense, (20a) should be judged false, and (20c) true. For test cases, the rationale is that if *had-to/moest* can scope over tense, participants should accept (20b) (contra *seem/leek*). If *had-to/moest* can scope below tense, they should accept (20d) (like *seem/leek*).

- (20) Context: Al has been a prime suspect for a crime that occurred last night in Montreal. Up to now, all of the evidence pointed to him being in Montreal last night. But just now, the detective receives fresh evidence that proves that Al was in fact in DC last night.

The detective says:

- | | |
|--|---|
| a. “Al seemed to be in DC last night” | (expected: false) |
| b. “Al had to be in DC last night” | (epi > tense: true, tense > epi: false) |
| c. “Al seemed to be in Montreal last night” | (expected: true) |
| d. “Al had to be in Montreal last night” | (epi > tense: false, tense > epi: true) |

Van Dooren finds that in English, participants judge *had-to*-sentences as true in 89% in (b) vs 6% in (d), the contrast we expect if *had-to* only scopes over tense. In Dutch, participants judge *moest*-sentences as true in 74% in (b) vs 46% in (d). She takes this to show that, both Dutch *moeten* and English *have-to* can scope over tense. Note that while she doesn't discuss it, the difference seems lower for Dutch than for English—which could reflect that the interpretation *tense > epi* is (more) available in Dutch. However, a problem is that she finds unexpected results for the baselines *seem/leek*—namely, 84% of acceptance for English *seemed* in (a) (expected false), and only 47% in (c) (expected true); for Dutch, she obtains 46% vs. 74%. Van Dooren speculates that this relates to focus, and comes from some participants “ignoring” past tense morphology to focus on evaluating the truth of the information provided by the embedded clause. Alternatively, her results may come from participants being free to assess *seemed* to any time point in the past. To resolve this issue, in our experiment, we added “to me” to ensure that *seemed* and *thought* were attributed to the same speaker.

3.3 Testing the predictions

The three theories we described in Section 3.1., combined with the less than clear results from van Dooren (2020a) in Section 3.2. give rise to several questions. Do epistemic modals always scope over tense or below tense? Are there languages where both options are available? How much variation is there between languages and what explains it? Do they come from differences in modals' syntax?

The subtle nature of these judgments, added to the potential competition with root interpretations (especially in the case of English *have-to*), makes the question hard to reliably elicit through fieldwork, or to settle based on corpus observations: we need scenarios with explicit contrasts between the two readings, and clear baselines ensuring that potential unacceptability does not arise from the intrinsic peculiarity of these scenarios. The experiments we present in the next sections build on van Dooren's (2020a) paradigm, crucially adding French as a test case.

While van Dooren does not discuss French, we take that her prediction would be for French *devoir* to pattern like English *have-to*. Indeed, French modals are usually taken to be (semi)-auxiliaries (Borgonovo & Cummins 2007; Hacquard 2010).¹⁶

¹⁶ As for *have-to* (see footnote 15), the term *semi-auxiliary* is commonly used for French modals, yet rarely explicitly defined. Here again, while *devoir/pouvoir*'s syntactic behavior is distinct from standard French auxiliaries *avoir* 'have' and *être* 'be', they still differ from lexical verbs. In French, this can be seen through various diagnostics, notably that they allow restructuring phenomena such as passivization with infinitives and clitic climbing (i)/(ii), and they give rise to Actuality Entailments (Hacquard 2006). This teases them apart from other predicates taking clausal complements like *vouloir*, which do not allow clitic climbing, passivization with infinitives; and precludes actuality entailments.

(i) Je **dois/veux** aider Jean.
 I must/want.1SG.PRS help.INF John
 'I must help John'

We ran the same experiments in Dutch (D), English (E) and French (F), asking participants to judge the acceptability of sentences like “It had to be raining”, presented in contexts that made either the epi > tense or the tense > epi interpretation warranted. We compare the judgments on *had-to/moest/devait* to *seemed/leek/semblait*, which we assume scopes under tense. We ran two versions of the experiment: in the first version (Experiment 1: offline ‘reflective’ judgments), participants had unlimited time to answer; in the second version (Experiment 2: online ‘speeded’ judgments), they were asked to answer as quickly as possible.

4 Experiment 1: ‘reflective’ judgments

The goal of Experiment 1 was to test the available interpretations (epi > tense vs tense > epi) for past epistemic modals, comparing English *had-to*, Dutch *moest*, and French *devait*, using identical material and methods. We build on van Dooren’s (2020a) design, notably in the use of the *seemed* vs. *had-to* comparison, and the presentation of sentences as direct speech, but crucially, add French.¹⁷

4.1 Participants

60 native speakers per language were recruited (Dutch, English, French) (30 participants per Lemma condition (*had-to/seemed*), counterbalanced across four lists).¹⁸ English participants were recruited via Amazon Mechanical Turk (www.mturk.com), Dutch and French participants via Prolific (www.prolific.com) (mean age: D: 26.5; E: 37.6; F: 28.2; D: 20 females; E: 19 females; F: 23 females). Two participants were excluded based on a native speaker check presented at the beginning of the experiment (1 Dutch, 1 French). An additional 25 participants were excluded because of low accuracy on fillers in the AJT and the BetT, as detailed below (D: 8, E: 7, F: 10). We therefore report results for 153 participants (D: 51, E: 53, F: 49).

4.2 Materials and design

Critical sentences were of the form “It had to/seemed to me to P,” with two possible continuations: “...but I **thought** that Q” or “...but now I **know** that Q.” P and Q were chosen to be mutually exclusive (for instance, P: “be raining yesterday” vs Q: “be snowing yesterday”). We call the two continuation contexts Cont-Past and Cont-Pres, respectively. The four sentence types are spelled out in (21). (22) provides English’s (21) equivalents in Dutch and French.

-
- (ii) Je le **dois/*veux** aider.
 I CLIT must/want. 1SG.PRS help.INF
 ‘I must help him’

¹⁷ Further differences are that van Dooren presents critical sentences with a context using a truth-value judgment task; we present no context before the critical sentence (beyond ‘the detective says:’), but manipulate the sentence’s continuation, asking for ‘sensible’/‘nonsensical’ judgments.

¹⁸ Our exclusion criteria were based on native language and age (above 18-year-old).

- (21) Cont-Past (Continuation indicates past mistaken belief that Q):
- a. It **seemed to me** to be snowing, **but I thought** that it was raining. (expected: ✕)
 - b. It **had to** be snowing, **but I thought** that it was raining. (epi > tense: ✓, tense > epi: ✕)
- Cont-Pres (Continuation indicates present certainty that Q):
- c. It **seemed to me** to be snowing, **but now I know** that it was raining. (expected: ✓)
 - b. It **had to** be snowing, **but now I know** that it was raining. (epi > tense: ✕, tense > epi: ✓)
- (22) Critical sentences:
- a. *English*: It {had to/seemed to me} P, {but now I know/but I thought} that Q.
 - b. *Dutch*: Het {moest/leek me} P, {maar nu weet ik/maar ik dacht} dat Q.
 - c. *French*: Il {devait/me semblait} P, {mais maintenant je sais/mais je pensais} que Q.

The continuation context **Cont-Past** (“but I thought Q”) allows us to test the availability of the epi > tense interpretation. “But I thought Q” describes a past (mistaken) belief that Q, which is compatible with a present epistemic claim that P, but not with a past epistemic claim that P. Therefore, if participants get the epi > tense interpretation for the modal verb, they should judge the full sentence ‘sensible.’ If they get tense > epi, they should judge the full sentence ‘nonsensical’ since it leads to a contradiction (P and Q cannot be true at the same time). In other words, **Cont-Past** forces the epi > Tense interpretation. Here, we expect *seemed* ((21a)) to be unacceptable, since *seem* scopes below tense.

The continuation context **Cont-Pres** (“but now I know Q”) allows us to test the availability of the tense > epi interpretation. “But now I know Q” describes a present certainty about Q, which is compatible with a past epistemic claim that P, but not with a present one. Therefore, if participants get the epi > tense interpretation, they should judge the full sentence ‘nonsensical’. If they get tense > epi, they should judge the full sentence ‘sensible’ (it does not lead to a contradiction). In other words, **Cont-Pres** forces the tense > epi interpretation. Here, we expect *seemed* ((21c)) to be acceptable (there is no contradiction).

We created 8 different critical sentences, listed in **Table 2**. Particular care was taken to ensure that the sentences make deontic interpretations unlikely. As discussed in the introduction, this is particularly important, since under a deontic interpretation, the modal is uncontroversially interpreted under tense.¹⁹ Note that for *seemed*, we add “to me” (respectively, “leek me” in

¹⁹ To ensure epistemic interpretations, we used stative prejacent and contexts in which a deontic interpretation is unlikely. This is also one of the reasons why in French, we test the *imparfait* rather than the *passé composé* (cf. 12a-b): since the aspectual morphology on *devoir* is constrained by the aspectual properties of its prejacent (Martin 2011; Laca 2012), and statives and perfective don’t naturally combine (the “Boundedness Requirement” of the perfective, Mari & Martin 2007; de Swart 1998), participants would have rejected “a dû” sentences for reasons unrelated to scope.

Dutch, and “me semblait” in French) to make sure that *seemed* and *thought* are evaluated by the same individual (if they are evaluated by different individuals, “seemed P... but I thought Q.” is non-contradictory).²⁰

	Sentence LEMMA = {<i>had-to/seemed to me to</i>} P	Continuation but {<i>I thought/now I know</i>} that Q
1	There LEMMA be more than 100 people at the party last night	there were fewer than 60.
2	The AC LEMMA be broken in the rental where we stayed	someone hadn’t turned it on correctly.
3	The barking dogs LEMMA be afraid of something last night	their owner simply returned home.
4	Tyler’s phone battery LEMMA be dead after the long hike	it was full.
5	The road LEMMA be slippery this morning	that was not the case.
6	The cat we got from the shelter LEMMA be starving	it had eaten too much.
7	It LEMMA be snowing this morning	it was raining.
8	The driver LEMMA be at fault in last night’s accident	he wasn’t.

Table 2: Critical sentences in English.

For fillers, we used sentences either sounding contradictory for no-controls (e.g. “Aron is American, and indeed, he is Italian”, expected to be judged ‘nonsensical’) or not contradictory for yes-controls (e.g. “Aron is American, and indeed, he was born in New York,” expected to be judged ‘sensible’).²¹ We had 8 fillers with yes/no variants. All sentences (critical cases and fillers) in Dutch, English and French are available in **Appendix A**.

Continuation context (Cont-Past vs Cont-Pres) was tested within subjects, Lemma (*had-to/seemed*) and Language (Dutch/English/French) between subjects.

²⁰ As pointed out by an anonymous reviewer, while solving the issue of anchoring the individual, this potentially introduced another issue, since it adds a disambiguation to the *seemed*-sentences that is not available for *had-to*.

²¹ Note that no-fillers are not strictly speaking contradictions, but pragmatically inappropriate. Crucially, “A, and indeed B” signals a relation of reinforcement between A and B, where B is expected to support A. These sentences are therefore expected to be judged nonsensical.

4.3 Procedure

The experiment was run online using the IBEX platform (Drummond, n.d.).²² The experimental material and procedure were identical in Dutch, English and French. After instructions (**Appendix B**), participants had a short training (3 examples with feedback). In the test phase, they were asked to judge the acceptability of 16 sentences (8 fillers, 8 critical cases), presented in randomized order. They first had to indicate, whether the detective uttering the sentence was speaking in a ‘nonsensical’ vs. ‘sensible’ way (*Acceptability Judgment Task*, AJT), and second how they would distribute a \$100 bet on what *really* happened, P or Q (*Betting Task*, BetT). Examples of the display are given in **Figure 1** (critical case with *had-to*) and **Figure 2A/B** (yes- and no-fillers). For the bet, participants could distribute their bet going from \$100 on P and \$0 on Q, to \$75 on P and \$25 on Q, \$50 on P and \$50 on Q, \$25 on P and \$75 on Q, and \$0 on P and \$100 on Q. The Betting Task further allowed us to increase the contrast between P and Q, indicating that they could not both be true at the same time. The placement of the P/Q options (right/left part of the screen) was counterbalanced across participants.

The detective says:

"It had to be snowing this morning, but now I know that it was raining. "

The detective is speaking in a...

○
nonsensical way

○
sensible way

*If you had \$100 to bet on what **really** happened, how would you distribute your money?*

It was snowing

○

100/0

○

75/25

○

50/50

○

25/75

It was raining

○

0/100

→ Click here to continue

Figure 1: Example of a critical trial, Experiment 1 (English, Cont-Pres).
Expected answer: AJT: ‘nonsensical way’; BetT: 50/50.

²² IBEX (short for Internet Based EXperiments) is an open-source platform designed for conducting psycholinguistic experiments online developed by Alex Drummond.

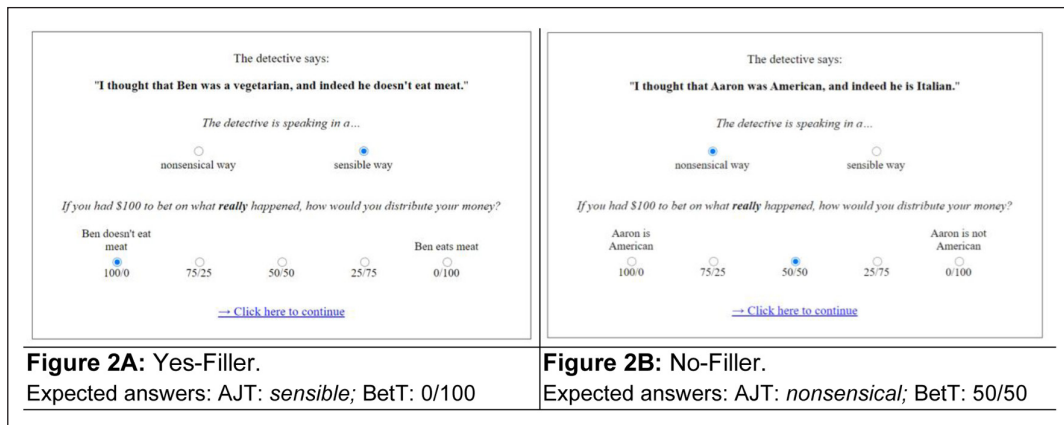


Figure 2: Examples of Yes-/No-fillers, Experiment 1 (English).

4.4 Rationale and expectations

For *had-to* /*moest/devait*), we test the hypotheses that: (H1) in all languages, epistemic can scope above tense (Hacquard 2006; van Dooren 2020b; contra Rullmann & Matthewson 2018); (H2) epistemic modals like Dutch *moeten* are modal verbs, therefore can both scope above tense (the tense of their complement), and below tense (the tense of their own clause). By contrast, English *have-to* and French *devoir* are (semi-)auxiliaries and only scope above tense (van Dooren 2020b). We compare judgments on *had-to* to judgments with the verb *seemed* (*to me*), which we assume is interpreted below tense in the three languages (*epi > tense, tense > epi). **Table 3** summarizes our expectations.

Acceptability Judgment Task (AJT). In the **Cont-Past** condition (i.e., + thought continuations), we expect participants to reject *seemed*-sentences, **Table 3**, row (a)), as *seemed* is interpreted below tense in the three languages ((a) should sound contradictory, since P and Q are incompatible). For test cases (*had-to*/*moest/devait*), if (H1) is true, participants in all languages should accept sentences (**Table 3**, row (b)). In **Cont-Pres** (i.e., + know continuations), we expect participants to accept *seemed*-sentences, **Table 3**, row (c)) (no contradiction). For test cases, if (H2) is true, only Dutch participants should accept these sentences, whereas English and French participants should reject them (**Table 3**, row (d)).

Betting Task (BetT). We use the Betting Task both as a control and exclusion criteria (for fillers, we excluded participants who made more than 2 mistakes out of the 4 ‘sensible’ fillers), and as an indicator of the actual interpretation of test sentences (as it shows whether participants think that P or Q is true). Note that we didn’t give specific instructions for nonsensical utterances on the Betting Task: participants could either use the 50/50 option or pick at random.

Condition	Sentence	Continuation	Accept. Judg. (AJT) Expectations			Betting Task (BetT) Expectations	
			D	E	F	If AJT = ok	If AJT = X
Cont-Past	a. It seemed to me P	but I thought Q	X	X	X	<i>Not expected</i>	P/Q
<i>epi > tense?</i>	b. It had to P	but I thought Q	ok	ok	ok	P	P/Q
Cont-Pres	c. It seemed to me P	but now I know Q	ok	ok	ok	Q	P/Q
<i>tense > epi?</i>	d. It had to P	but now I know Q	ok	X	X	Q	P/Q

Table 3: Sentences and expected answers if following van Dooren (2020b) for (i) the *Acceptability Judgment Task* ('ok' = 'sensible', X = 'nonsensical'); (ii) the *Betting Task*. For instance, P = "It was raining.", Q = "It was snowing."

4.5 Results

4.5.1 Participant exclusion

The accuracy on both yes- and no-fillers on the AJT was high in all languages (general mean accuracy: D: 97.7%; E: 92.1%; F: 91%). We excluded 19 participants who had <75% accuracy (more than 2 errors out of 8 fillers) on the AJT (10.6%) (D: 3, E: 7, F: 9). To compute correct answers on the BetT (where we consider only yes-fillers, since there is no expected good answer for no-fillers), we merge answers 0 and 25 together as 'P', and 75 and 100 as 'Q' (i.e. we consider as 'correct' answering 25 or 0 for [0], and answering 75 or 100 for [100]). Accuracy on the BetT was high in all languages (D: 84.2%; E: 97.1%; F: 86.8%). We excluded 7 participants (D: 5, E: 0, F: 2) who had <75% accuracy (i.e., more than 2 mistakes on the bet, out of the 4 yes-fillers).

4.5.2 Analysis

Data analyses were conducted using R (R Core Team 2013), using the package lme4 (Bates et al. 2014; 2015). We first analyze each language separately, then compare between languages.

*Analysis per language (Interaction Lemma*Context).* To analyze participants' responses for each language, we fitted a generalized linear mixed-effects model with a maximal random effect structure (following Barr et al., 2013). The dependent variable was Answer ('sensible' vs 'nonsensical', binomial), with Lemma (*had-to/seemed*) and Context (Cont-Past/Cont-Pres) and their interaction as fixed effects. Categorical predictors (Lemma, Context, Language) were sum-coded. Random effects included Subject by Context and Item by Context and Lemma. We compare the full model to a reduced model using a likelihood ratio test (glmer syntax of the full model: Answer = 1 ~ Lemma*Context + (Context|Subject) + (Context + Lemma|Item); reduced model: Answer = 1 ~ 1 + (Context|Subject) + (Context + Lemma|Item)). Answers were coded as 1 ('sensible') and 0 ('nonsensical'). To fix convergence issues, we used the "bobyqa" optimizer in R (glmerControl(optimizer = "bobyqa", calc.derivs = FALSE)). To further investigate the significant Lemma*Context interaction, we then used estimated marginal means (EMMs) with the emmeans package in R (Lenth 2024), with p-values computed using the Satterthwaite approximation as implemented in the emmeans package.

Comparison between languages. To compare responses between languages, we test for the interaction between Language and Context, with Answer as dependent variable (binomial). We run analyses on *seemed* and *had-to* separately. The full model includes fixed effects of Language, Context and their interaction, with random intercepts and slopes for Context by subject, and random intercepts and slopes for Context and Language by Item (glmer syntax: Answer = 1 ~ Language*Context + (Context|Subject) + (Context + Language|Item)). We compare this to reduced models without the interaction (Answer = 1 ~ Language + Context + (Context|Subject) + (Context + Language|Item)) using a likelihood ratio test.

4.5.3 Results

Figure 3 displays the mean proportion of ‘sensible’ answers by conditions for Dutch, English and French on the AJT (see **Appendix C, Table C1**). We see that in the **Cont-Past** condition, participants in all languages tend to reject baseline *seemed/leek/sembloit* sentences, though at slightly higher rates than expected ((a): D: 23%; E: 24%; F: 24%). By contrast, they tend to accept *had-to/moest/devait* sentences—at least, more often than *seemed* sentences ((b): D: 55%; E: 40%; F: 57%). In **Cont-Pres**, participants in all languages tend to accept *seemed/leek/sembloit* sentences, as expected ((c): D: 85%; E: 80%; F: 91%). For *had-to/moest/devait*, participants in English and French tend to reject the sentences—at least, more often than the same sentences with *seemed*, whereas in Dutch, the difference between *moest* and *leek* is smaller ((d): D: 79%; E: 52%; F: 70%). Results from the model comparison testing for the Lemma*Context Interaction are reported in **Table 4**, with associated β -estimate, z-values and p-values of the full model. Simple effects of Lemma based on EMMs are summarized in **Table 5**.

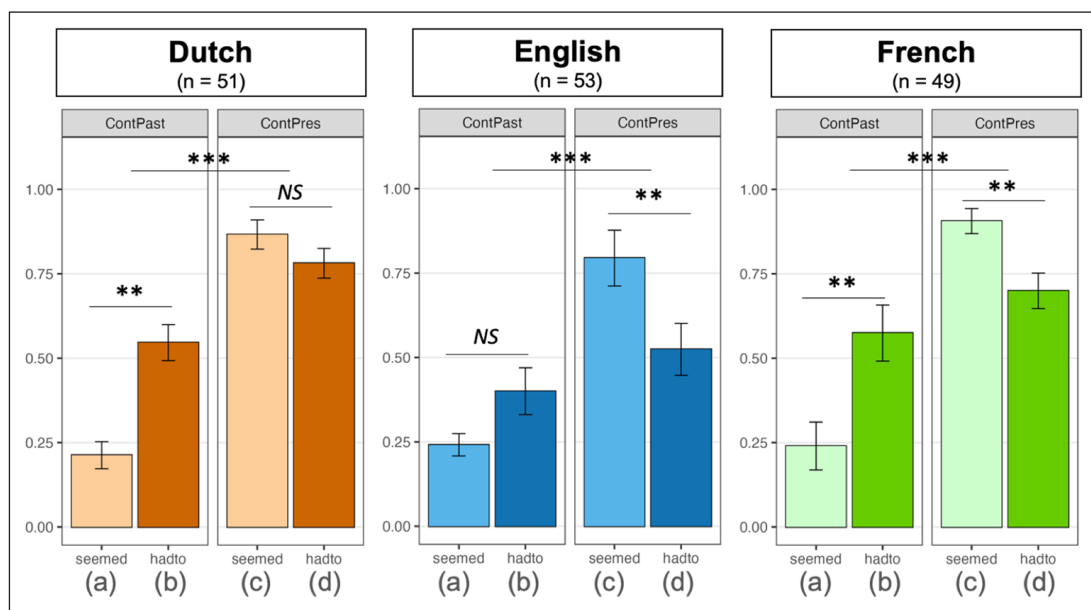


Figure 3: Proportion of ‘sensible’ responses by condition for Dutch (orange), English (blue) and French (green) for *seemed* (light) and *had-to* (dark), AJT, Experiment 1 (n = 153).

	Result of the LRT (model comparison)	Estimate (β), Std. Error, z-value, p-value
Dutch	$\chi^2(1) = 8.36, p = .0038^{**}$	$\beta = -1.50, SE = 0.47, z = -3.19, p = .001$
English	$\chi^2(1) = 15.21, p < .001^{***}$	$\beta = -1.45, SE = 0.34, z = -4.31, p < .001$
French	$\chi^2(1) = 22.03, p < .001^{***}$	$\beta = -2.06, SE = 0.40, z = -5.12, p < .001$

Table 4: Result of the model comparison testing Interaction Context*Lemma with associated Estimate (β), Std. Error, z-value, p-value for m1 by language, Experiment 1 (full model m1: Answer = 1 ~ Lemma*Context + (1 + Context|Subject) + (1 + Context + Lemma|Item)).

	Cont-Past				Cont-Pres			
	β	SE	z	p-value	β	SE	z	p-value
Dutch	-2.07	0.54	-3.83	0.0001 ^{**}	0.94	0.87	1.09	0.27 (NS)
English	-0.91	0.47	-1.91	0.056 (NS)	1.99	0.61	3.26	0.0011 ^{**}
French	-2.44	0.68	-3.60	0.0003 ^{**}	1.68	0.59	2.84	0.0045 ^{**}

Table 5: Simple effects of Lemma (*seemed* vs. *had-to*) within each Context by Language, based on estimated marginal means (EMMs) from the full model (m1), Experiment 1.

Comparison between Languages. For *seemed/leek/semblait*, model comparison reveals no significant interaction between Language*Context ($\chi^2(2) = 2.82, p = 0.244$), indicating that participants across the three languages treat *seemed*-sentences similarly. Looking at simple effects, only the main effect of Context is significant ($\beta = 3.35, SE = 0.36, z = 9.42, p < .001$) (reflecting that participants overall accept more sentences in Cont-Pres than Cont-Past). For *had-to/moest/devait*, likewise, there is no significant Language*Context interaction ($\chi^2(2) = 2.04, p = .36$). Looking at simple effects, the model shows a significant effect of Language ($\beta = 1.32, SE = 0.53, z = 2.51, p = .012^*$; $\beta = 1.06, SE = 0.45, z = 2.34, p = .019^*$), indicating lower acceptance rates for English *had-to* than French and Dutch. Results of the models' output are available in **Appendix G**.

Figure 4 shows results by subject. Each dot corresponds to a participant (a jitter was added in order to make the results readable). We see that *seemed* participants (grey dots) are distributed across the bottom part of the graph (right to the diagonal). *Had-to* participants (black dots) are spread, and do not show a clear pattern. This is interesting, since one possibility was that the tense > epi interpretation is accepted only by some speakers. This doesn't seem to be the case. Details of the results by context and sentence are available in **Appendix C**.

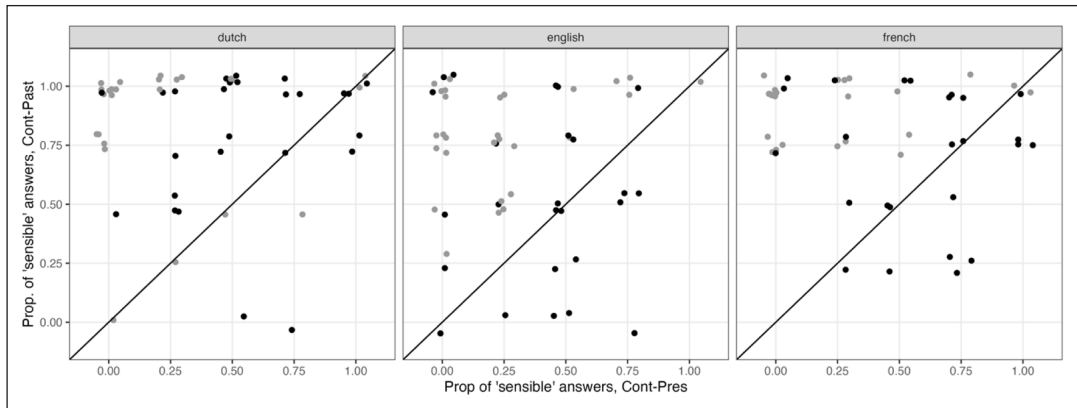


Figure 4: Answers by subject for *seemed* (grey) and *had-to* (black) (*seemed*: 77 participants across the 3 languages; *had-to*: 76 participants). Each dot corresponds to one participant. The y-axis corresponds to the proportion of ‘sensible’ answers given for **Cont-Past** (out of 4 trials), the x-axis, the proportion for **Cont-Pres**. Participants who always reject sentences with *had-to* would appear in the left bottom corner (0;0); participants who always accept it, in the right up corner (100:100).

Figure 5 shows the distribution of bets on ‘sensible’ answers.²³ We merged answers 0/25 together, and 75/100. We see that regardless of the lemma (*seemed/had-to*), when participants answer ‘sensible’ in **Cont-Pres**, they always pick the Q option (white bar).

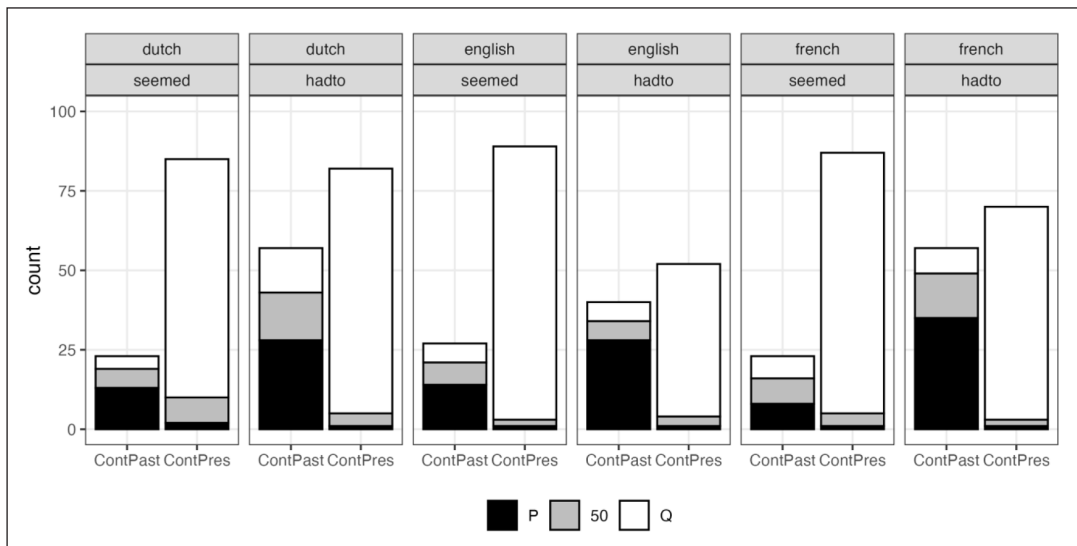


Figure 5: Distribution of bets based on condition (raw counts, only out of ‘sensible’ answers), Experiment 1.

²³ Results for ‘nonsensical’ answers (where participants were supposed to pick at random) are available in Appendix D.

4.6 Interim discussion

In the **Cont-Past** condition, we find higher acceptance rates for *had-to/moest/devait* in all languages, when compared to *seemed/leek/semblait*. This is what we expect if these sentences can be interpreted as a present epistemic claim about a past eventuality (epi > tense) (contra *seemed/leek/semblait*). This is in line with predictions from Hacquard 2006; Hacquard & Cournane 2016; van Dooren 2020b; contra e.g. Rullmann & Matthewson 2018). In **Cont-Pres**, conversely, we find both in English and in French lower acceptance rates for *had-to/devait* than for *seemed/semblait*. This is what we expect if the epistemic modal cannot be interpreted as a past epistemic claim (*tense > epi). In Dutch however, the difference is not significant in this condition (*moest*: 78% vs *leek*: 87%), which is what van Dooren's (2020b) theory predicts: according to her, for Dutch, the two interpretations are available. It is worth highlighting there does not seem to be interspeaker variation: we don't find two groups of participants, some always accepting *had-to* and some always rejecting it in the **Cont-Pres** condition. A third important result, allowed by the direct comparison between languages, is that English *had-to* is judged significantly less natural than its Dutch and French counterparts (app. -15/20%). This result aligns well with Rullmann & Matthewson's (2018) point that *had-to* in sentences like (5)/(6) are disliked by native speakers (**Section 2.1**). We think specificities of English *have-to* can help clarify part of the debate. We will come back to this in the general discussion, where we discuss the idea that the lower acceptability of *had-to* comes from specific licensing conditions on *have-to* (Williams et al. in prog.).

While we do find differences between *had-to* and *seemed*, our results are not as clear cut as one might expect. In particular, the acceptance rates we find for *had-to* sentences in **Cont-Pres**, both in English and French, are quite high: around 52% in English, and 70% for French, in situations where we expect participants to simply reject the sentence. Second, we also find quite high acceptance rates for *seemed* in **Cont-Past** (sentences like "It seemed to P but I thought that Q"), with average acceptability ratings around 25% in all language, while we expected participants to reject it. Why is that so?

Starting with the latter, it may be that participants interpret the two epistemic claims, *seemed* P and *thought* Q, at different past times. If so, the claims are not incompatible anymore. However, we still find a consistent pattern across the three languages—showing that the design of our controls is an improvement on van Dooren (2020a).

Then, why do participants accept *had-to* in **Cont-Pres**? Several factors could play a role. It could be that our contexts and design did not make the incompatibility between P and Q salient enough: for instance, one could construct worlds where snow and rain might happen at the same time. However, in 3 cases, there was an explicit contrast between 'P' and 'not P' (sentences 1, 5 and 8 in **Table 2**); we would then expect differences between these and the other contexts, which we don't seem to find (see **Appendix C**). Moreover, the betting task helped partitioning worlds, by forcing to bet on P- against Q-worlds. A second possible explanation is that some participants

(re)interpret *had-to/devait* sentences as expressing deontic modality (e.g. “had-to P” as ‘based on *past* rules, it was necessary that P’, which does not contradict *now knowing* Q). Again, we don’t think this is the explanation: we built the material to avoid this option (footnote 19), making sure that the deontic was extremely unnatural if not impossible (e.g. “The AC had to be broken” or “The driver had-to be at fault”). Finally, despite our design avoiding any overt narrative context (sentences were presented as direct speech, and set in the present tense), participants might read the utterances as FID, read this as a detective novel set in the past tense: “It had to be snowing, he thought.”

The original goal of Experiment 2 was to try to get to the “primary” interpretations and prevent potential reinterpretation strategies by asking participants to answer under time pressure. While we now believe that our design was not well suited to test for this, the study still serves as a (partial) replication of Experiment 1, testing the robustness of the findings.

5 Experiment 2 (‘speeded’ judgments)

5.1 Goal

In Experiment 1, participants were encouraged to take their time to make the judgments, which we hypothesized may have pushed them to reinterpret the sentences—either with a deontic interpretation or imagining a past narrative context (FID). The (initial) goal of Experiment 2 was to prevent such reinterpretation strategies by asking participants to answer quickly. The paradigm also allowed us to record participants’ response time, which we couldn’t analyze in Experiment 1 since the AJT and the BetT appeared on the same screen.²⁴

Manipulating time pressure has been used as one way to study the online processing of various phenomena, notably potentially “ambiguous” sentences. It has been used to show, for instance, that there is a “cost” associated to scalar implicature generation (see e.g. Bott & Noveck 2004; Noveck & Posada 2003).²⁵ Here—if we assume that FID is not the “default” reading, in absence of past narrative discourse context, it should come with a “cost”: we therefore expect, if the high rates found for *had-to* in **Cont-Pres** in Experiment 1 reflect FID, to obtain lower acceptance rates in *had-to* in **Cont-Pres** when time pressure is added.

²⁴ Experiment 2 was more exploratory. Note that there are few experimental studies about the online processing of FID (experimental work exists, but focuses on for instance, whether and how readers detect the protagonist’s perspective, see Bray 2007; Kaiser 2015; Meuser 2022). Here, we initially made the assumption that FID came at a cost, but as discussed with a reviewer, this assumption may not be justified. However, approaches to FID which postulate a covert syntactic operator (e.g. Hacquard 2006; Charnavel 2025) may support this original assumption.

²⁵ The cost associated to scalar implicature computation is often taken to reflect pragmatic processes, but in itself, a first limitation is that the paradigm does not indicate the nature of the cost: it could be pragmatic as well as syntactic processes. We considered other methodologies like dual tasks (used to reduce the amount of cognitive resources; see e.g. De Neys & Schaeken 2007). We chose to manipulate time pressure to stay close to Experiment 1.

5.2 Participants

180 participants were recruited using Prolific (www.prolific.com) (D: 61; E: 58; F: 61; 8 groups for counterbalancing) (mean age: D: 31.3; E: 41.3; F: 32.4; D: 25 females, 35 males, 1 non-binary; E: 29 females, 26 males, 1 non-binary, 2 unknown; F: 25 females, 34 males, 1 non-binary, 1 unknown). We excluded 1 (Dutch) participant who participated twice in the experiment, 29 participants because of low accuracy (<75%) on fillers on the AJT (D: 4, E: 20, F: 5) and 4 participants for low accuracy (<75%) on comprehension questions (D: 1, E: 2, F: 1). We therefore report results for 148 participants (D: 55, E: 38, F: 55).²⁶

5.3 Procedure

All experiments were coded using PennController for Ibex (Zehr & Schwarz, 2018) (<https://www.pcibex.net/>). We used the same material as in Experiment 1. The main change was that participants were now asked in the instructions to answer as quickly as possible: instead of “Please take your time in completing this task, you may want to read them a second time or take a break in between”, they saw: “IMPORTANT // Time is limited, so answer as quickly as possible! // Just make sure you’ve read the whole sentence before answering”. Like in Experiment 1, after instructions and a short training, they had 16 sentences to judge (8 test, 8 fillers), presented in randomized order. On a first screen, they had to indicate whether the sentence was ‘nonsensical’ vs. ‘sensible’ (AJT, *i* in **Figure 6**); then on a second screen, after clicking a fixation cross, they had to indicate what they thought *really* happened, ‘P’ or ‘Q’ (BetT, *ii* in **Figure 6**). This allowed us to record their response time.

We also modified the way the sentence appeared: while in Experiment 1, the whole sentence was presented at once, in Experiment 2, it unfolded incrementally from left to right, the first part of the sentence over the other, at a rate of 40ms/letter (i.e., on average 3500ms for the full sentence to appear). We counterbalanced for the placement of the answers: half of the participants saw ‘sensible’ on the right and ‘nonsensical’ on the left, the other half saw ‘sensible’ on the left and ‘nonsensical’ on the right. Because we wanted to simplify the display, we removed the fictional detective speaker,²⁷ and changed the Betting Task for a binary choice: rather than having to pick between 5 options to make a bet, participants simply had to click on the true

²⁶ The accuracy on the AJT fillers was overall high (D: 94.6%; E: 84.3%; F: 93.2%). However, for reasons that remain unclear, English participants in Experiment 2 performed quite poorly on three fillers: “Al had two children, and indeed he has only one child” (69%); “there was a mosquito in the room, and indeed it was just a fly” (66%); “I had forgotten my umbrella, and indeed I had it with me” (69%). We still applied the same exclusion criteria as for Experiment 1, excluding participants who had <75% accuracy, but end up with only 38 participants in the English group.

²⁷ We now think this was an damageable decision, we should have kept the detective and quotes.

statement, with the two options P or Q displayed in two boxes. At the end, participants were asked for comments about the task. Examples of the experiment are available at <https://farm.pcibex.net/r/TLMTmo/> (English), <https://farm.pcibex.net/r/vCyMpc/> (Dutch) and <https://farm.pcibex.net/r/ffkCsV/> (French). As in Experiment 1, Continuation (+ *know*/+ *thought*) was tested within subjects, Lemma (*had-to/seemed*) and Language (Dutch/English/French) between subjects.

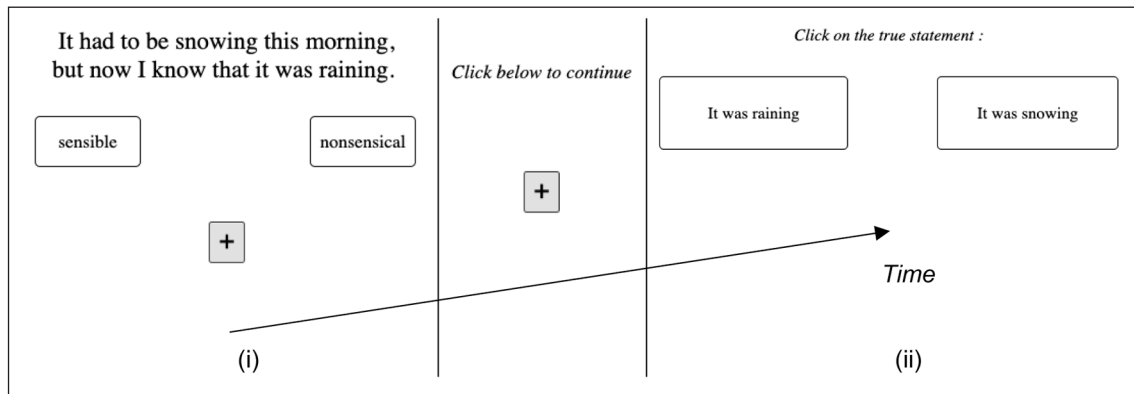


Figure 6: Display for Experiment 2. (i): AJT; (ii): BetT.

Assuming that reducing the amount of time available to make the judgment prevents participants from reinterpreting the sentence (e.g. deontic or FID), we expect to reduce the amount of answers ‘sensible’ in **Cont-Pres** for *had-to* for French and English, but not Dutch. We also expect to reduce acceptance rates for *seemed* in **Cont-Past**. Looking at response times, we expect participants to take longer to answer ‘sensible’ than ‘nonsensical’ in **Cont-Pres** for *had-to* sentences for French and English.

5.4 Results

5.4.1 Analyses

We first ran the same analyses as for Experiment 1 (see Section 4.5.2). To compare results between the two experiments, we fitted a generalized linear mixed-effects model with a maximal random effect structure, with Answer as dependent variable (binomial) and Experiment (Experiment1: ‘offline’ vs Experiment2: ‘online’) as fixed effect. Random effects include Subject by Context and Item by Context, Lemma and Experiment. We compare the full model to a reduced model using a likelihood ratio test (full model: $\text{Answer} = 1 \sim \text{Experiment} + (\text{Context}|\text{Subject}) + (\text{Context} + \text{Lemma} + \text{Experiment}|\text{Item})$; reduced model: $\text{Answer} = 1 \sim 1 + (\text{Context}|\text{Subject}) + (\text{Context} + \text{Lemma} + \text{Experiment}|\text{Item})$). (Unconclusive) results of Response Times (RTs) analyses are reported in **Appendix E**.

5.4.2 Results

Figure 7 displays the mean proportion of ‘sensible’ answers by conditions for Dutch (D), English (E) and French (F) in Experiment 2 (**Appendix C, Table C2**). As in Experiment 1, we see that participants tend to reject *seemed* in **Cont-Past** (i: D: 23%; E: 18%; F: 28%), and accept it in **Cont-Pres** (ii: D: 94%; E: 78%; F: 91%). For *had-to*, we find higher acceptance rates for *had-to* than *seemed* in **Cont-Past** (iii: D: 62%; E: 44%; F: 60%), and lower than *seemed* in **Cont-Pres** (iv: D: 76%; E: 69%; F: 83%).

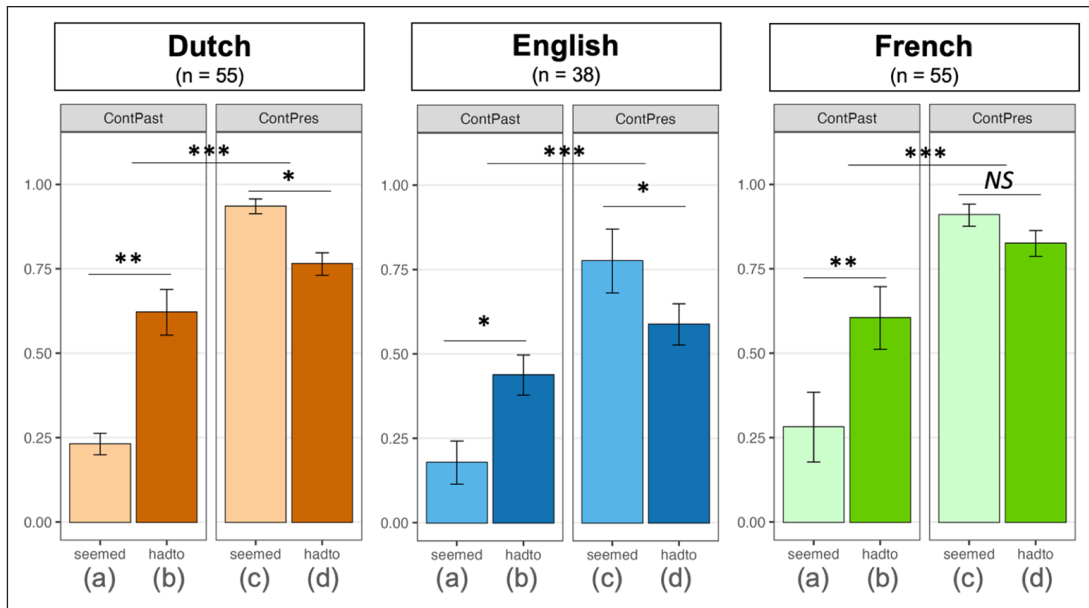


Figure 7: Proportion of ‘sensible’ responses by condition for Dutch (orange), English (blue) and French (green) for *seemed* (light) and *had-to* (dark), AJT, Experiment 2 (n = 148).

5.4.3 Analysis

Analysis per language. Results from the model comparison testing for the Lemma*Context Interaction are reported in **Table 6**, with associated β -estimate, z-values and p-values of the full model. As in Experiment 1, we find significant Lemma*Context interactions in all languages (Dutch: $\chi^2(1) = 18.2$, $p < .001^{***}$, English: $\chi^2(1) = 11.9$, $p = 0.00056^{***}$; French: $\chi^2(1) = 15.2$, $p < .001^{***}$). Further analysis using EMMs reveal significant simple effects of Lemma in **Cont-Past** for all three languages, with significantly lower acceptance rates for *seemed* compared to *had-to* (in Experiment 1, the difference in English is only marginally significant) (Dutch: $\beta = -2.43$, $SE = 0.57$, $p < .0001$; English: $\beta = -2.24$, $SE = 0.89$, $p = .012$; French: $\beta = -2.43$, $SE = 0.68$, $p = .0003$). In **Cont-Pres**, Dutch and English showed significant higher acceptance rates for *seemed* relative to *had-to* (Dutch: $\beta = 1.86$, $SE = 0.84$, $p = .026$; English: $\beta = 2.15$, $SE =$

1.06, $p = .043$), while French showed no significant difference ($p = .174$) (in Experiment 1, the difference was significant in French but not in Dutch).

Results by subjects, by context and by sentence are available in **Appendix C**.

	Result of the model comparison	Estimate (β), Std. Error, z-value, p-value
Dutch	$\chi^2(1) = 18.17, p < .001^{***}$	$\beta = -2.15, SE = 0.47, z = -4.60, p < .001$
English	$\chi^2(1) = 11.90, p = 0.00056^{***}$	$\beta = -2.19, SE = 0.59, z = -3.74, p < .001$
French	$\chi^2(1) = 15.23, p < .001^{***}$	$\beta = -1.68, SE = 0.40, z = -4.24, p < .001$

Table 6: Result of the model comparison testing Interaction Context*Lemma with associated Estimate (β), Std. Error, z-value, p-value for m1 by language, Experiment 2 (full model m1: Answer = 1 ~ Lemma*Context + (Context|Subject) + (Context + Lemma|Item)).

	Cont-Past				Cont-Pres			
	β	SE	z	p-value	β	SE	z	p-value
Dutch	-2.43	0.571	-4.25	<.0001**	1.86	0.837	2.23	0.0260*
English	-2.24	0.887	-2.52	0.0117*	2.15	1.063	2.02	0.0432*
French	-2.43	0.678	-3.58	0.0003**	0.93	0.687	1.36	0.1742 (NS)

Table 7: Simple effects of Lemma (seemed vs. had-to) within each Context by Language, based on estimated marginal means (EMMs) from the full model (m1), Experiment 2.

Comparison between Languages. For *seemed*, as in Experiment 1, model comparison reveals no significant interaction between Language*Context ($\chi^2(2) = 0.0031, p = .99$): participants do not treat *seemed*-sentences differently across the three languages. Again, the main effect of Context is significant ($\beta = 3.27, SE = 0.47, z = 6.97, p = 3.3e-12$) (participants overall accepting more sentences in Cont-Pres than Cont-Past). For *had-to*, there was likewise no significant Language*Context interaction ($\chi^2(2) = 1.82, p = 0.40$). The model shows a main effect of Context ($\beta = 1.22, SE = 0.31, z = 3.97, p < .001$), and a significant effect of language ($\beta = 1.23, SE = 0.45, z = 2.77, p = .006^{**}$). Details of the models' outputs are available in **Appendix G**.

Comparison between Experiments. We find no difference between experiments, be it overall ($\chi^2(2) = 0.91, p = 0.34$) or looking at each subcondition. Results from likelihood ratio tests by subconditions are given in **Table F, Appendix F**.

Figure 8 shows the distribution of answers (on 'sensible' answers). Results for 'nonsensical' answers (where participants were supposed to pick at random) are available in **Appendix D**. As

in Experiment 1, when participants answer ‘sensible’ in **Cont-Pres** condition, they systematically pick the Q option.

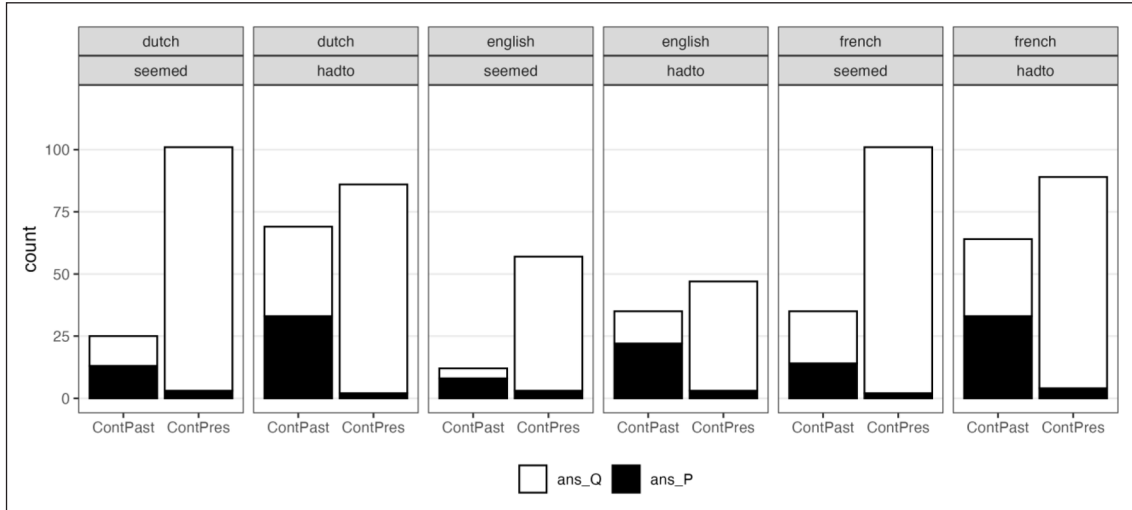


Figure 8: Distribution of answers (Q vs P) based on condition (raw counts, only out of ‘sensible’ answers), Experiment 2.

5.5 Discussion

In Experiment 1, we found quite high acceptance of *had-to* in **Cont-Pres**, which we hypothesized could be due to a “reinterpretation” of these sentences as FID. To explore this hypothesis, we implemented a “speeded” version of the Experiment, aimed at testing whether participants change their response when asked to answer quickly, and allowing us to record their Response Times.²⁸ The manipulation did not affect participants behaviors in the way we expected. As in Experiment 1, we find that English, Dutch, and French participants accept *had-to*/*moest*/*devait* in Cont-Past in higher rates than *seemed*/*leek*/*semblait*. Unlike Experiment 1, we find no significant difference between *seemed* and *had-to* in the critical **Cont-Pres** condition for French (Table 7), and a now significant difference for Dutch. This warrants discussion.

The effect in Dutch seems mostly due to the baseline, with higher acceptance of *leek* sentences in Experiment 2 (94%) than in Experiment 1 (87%) (compared to *moest*: Experiment 1 78%; Experiment 2: 76%). For French, if this pattern reflects a real trend, it may actually go against the idea that the tense > epi interpretation is “harder to access”. The results may suggest that when

²⁸ For follow-up studies, we think it is more insightful to provide participants with unlimited time. Indeed, as pointed out by a reviewer, speeded judgements on syntactic/semantic phenomena often reveal partial and ungrammatical interpretations that are not allowed by the structure and are only temporary—see cases of linguistic illusions (e.g., Wagers et al. 2009; Parker & Phillips 2016; Muller & Phillips 2020).

asked to answer quickly, participants in fact tend to accept *devait* slightly more often in **Cont-Pres**. The difference is not significant, but we expected the opposite pattern. Moreover, when we consider response times on this specific condition (see **Appendix E**), we see that participants are overall 700ms slower to reject than to accept the sentence (mean RT for ‘sensible’ answers: 4168ms; for ‘nonsensical’ answers: 4816ms), whereas we find the opposite tendency for *devait* in **Cont-Past** (participants who accept the sentence, as we expected them to, are 500ms slower than participants who reject it). So, if anything, it seems that the change in our design facilitates the tense > epi interpretation.

If we keep our initial assumption that FID comes with a cost, this means that the high acceptance rates in Experiment 1 are not due to FID. Alternatively, it could be that FID does not come at a cost, even in the absence of an overt past narrative context when participants have to enrich the context themselves.

Finally, another difference we introduced might partly explain the difference with Experiment 1. Indeed, in order to simplify the task, we removed the fictional detective and quotation marks. This change might have left participants more free when judging sentences like “It had to P, but now I know Q”, to introduce a different epistemic source for the first part of the statements, allowing them to anchor “It had to P” for instance to a report in the newspaper, while anchoring “but now I know Q” to themselves. As a result, there is no incompatibility between the two conjuncts. Similarly, acceptance of the “It had to P, but I thought Q” would not show that epistemics can scope under tense, as no potential incompatibility could arise if two separate epistemic sources are chosen for the first and the second part of the statement. This could also explain that participants tend to choose Q in this condition in the betting task (**Figure 8**).

6 General discussion

6.1 Interpreting the findings

The goal of our experiments was to assess three different theories of modal-temporal interactions. These theories make different predictions regarding the interpretation of past tense modals in sentences like “Al **had to** be in the bathroom” or “It **had to** be raining.” According to Hacquard (2006), universally, epistemic modals scope over tense. In sentences with a past tense modal, even though the past tense morpheme appears on the modal, it is interpreted in its scope, resulting in a present epistemic claim about a past event. Following her theory, in our experiment, participants in all languages should accept *had-to/moest/devait* in Cont-Past (“It had to be raining, **but I thought that** it was snowing”) more than *seemed/leek/semlait*, as the latter only allow past > epi. They should also reject it in Cont-Pres (“It had to be raining, **but now I know** that it was snowing”), where they should accept *seemed/leek/semlait*. Rullmann & Matthewson (2018), contrastingly, state that epistemic modals universally scope *below* tense. If this is right, in our experiment,

participants should reject *had-to/moest/devait* in Cont-Past, while accepting them in Cont-Pres. Van Dooren (2020b), finally, hypothesizes that the lexical status of the modal (verb or (semi-) auxiliary) determines its scopal interaction: Epistemic modal auxiliaries can only scope *over* tense, while epistemic modal verbs can scope *over* and *below* tense. This theory predicts cross-linguistic differences: assuming that English and French modals are (semi-)auxiliaries, English and French participants should *accept* *had-to/devait* in Cont-Past and *reject* them in Cont-Pres, whereas, assuming that Dutch modals are verbs, Dutch participants should accept them in both Cont-Past and Cont-Pres.

What have we learned by comparing English, Dutch and French? We identify three key results. First, in both experiments, in Cont-Past, we find that participants in all languages accept sentences with *had-to/moest/devait* (significantly more often than *seemed/leek/semblait*). This suggests that that *epi > tense* is available in all three languages, in line with predictions from both Hacquard (2006) and van Dooren (2020b). Recall that this contrasts with judgments for root modal sentences, such as “Al had to do the dishes”—which uncontroversially express past obligations (*past > modal*) and not present ones (**modal > past*). Second, in Cont-Pres, we find in Experiment 1 that both English and French participants accept *had-to/devait* less often than *seemed/semblait*. This suggests that *past > epi* is not (or at least, less) available. In Dutch however, no such contrast is found, which is what we expect under van Dooren’s theory. The other two theories don’t predict differences between languages.

However, these first two findings are not so clear cut, and judgments are less crisp than expected. Though we do find the critical difference with *seem*, English and French participants in Cont-Pres still accept *had-to/devait* at quite high rates (55%/70%), which is hard to understand if *tense > epi* is simply unavailable. (i) Why do participants accept those sentences? Conversely, in Cont-Past, they accept *had-to/devait* at 40%/55% only. (ii) Why don’t they accept them more? What explains that in almost half of the cases, they judge them nonsensical?

Let’s begin with (ii). Why do so many participants reject “had-to P... but I thought Q”? We think this may be explained by independent pragmatic factors. Without any context, “but I thought” continuations sound weird: talking about past mistaken belief is less often a topic of conversation—i.e., less conversationally relevant—than describing the true state of the world. Thus, Cont-Past continuations may be likely to lead to ‘nonsensical’ answers, because they involve stating first a true fact then followed by a mistaken belief about it, which is conversationally odd.²⁹ One interesting follow-up would be to see whether reversing the order of the two clauses affects the judgments (e.g., “I thought that it was snowing, but (in fact) it had to be raining” vs “Now I know that it was snowing, but it had to be raining”), or, since the weirdness of these

²⁹ The fact that we used a within subject design for Context and between for Lemma might also have contributed to this.

examples seems to come from the lack of preceding context, to manipulate the Question Under Discussion so that talking about mistaken beliefs becomes relevant.

Let's turn to (i), which we see as more puzzling, and is a challenge to both van Dooren's and Hacquard's approaches. Why don't French and English participants just reject "had to P... but now I know"? First, it may be that the contrast between P and Q was not clear enough in some of our sentences (e.g., it can be raining and snowing at the same time). This might have been an issue especially in Experiment 2, where participants saw the sentence in isolation—whereas in Experiment 1, the Betting Task forced them to partition the world into P/Q-worlds. However, that cannot be the whole story, since some items contrasted P and not(P). A second possibility is that (some) participants created scenarios or stories akin to FID-discourse (as discussed in Section 2, examples (7)/(10)/(13)). Namely, they would imagine that the detective is going back in time and talks about an earlier epistemic event (of him falsely believing that P was true). Indeed, though we try in our design to minimize the odds of participants shifting to FID by presenting sentences in direct discourse (cf "The detective says:"), this might leave too many parameters open. Moreover, as noted e.g. by Kaiser (2015) (for theoretical work, see also Fludernik 1993, a.o.), linguistic expressions whose interpretation depends on reference to some kind of judge/evaluator (some kind of 'epistemic anchor') like epistemic modals, epithets, predicates of personal taste or appositives, are particularly prone to elicit FID. Her experiment 3 shows that readers indeed tend to switch to FID when sentences involve epistemic modals. Note that if participants' answers come from FID, we might expect to get patterns looking at variation between participants and/or items.³⁰ And indeed, amongst our 8 sentences, further inspection shows that some might lead to story-telling contexts more easily (for instance "The barking dogs LEMMA be afraid of something last night" vs "It LEMMA be snowing this morning").

Before getting further, we would like to make a small clarification note. While one could describe *had-to/devait/moest* sentences as "ambiguous," it is crucial to specify at what level of analysis. Here, at least two interpretations seem available: the 'direct discourse' one, and the 'FID' one. The difference between this "ambiguity" and Dutch would be that the 'epi > tense' reading that French/English participants access is (contra Dutch) not at the syntactic level proper, it is at the pragmatic one, whereas following van Dooren, the "ambiguity" of sentences is at the syntactic level. (Of course, FID is also available for Dutch speakers.).

Coming back to our main results, our third key finding is the overall lower acceptability for English *had-to*, as compared to Dutch *moest* and French *devait*.³¹ We believe this connects to the

³⁰ There is some experimental work on FID suggesting there is variation among speakers with respect to accessing a perspectival shift (Bray 2007; Mikulan & Legac 2014).

³¹ This, actually, aligns with Rullmann & Matthewson's (2018) point (see Section 2.1) that examples like (3) are disliked by English speakers. While Rullmann & Matthewson take this as evidence that epi > tense is unavailable, we believe that this comes from specific licensing conditions of *have-to*.

fact that in English, *have-to* is rarely used to convey epistemic modality—as evidenced by corpus studies, *must* has increasingly specialized for epistemic necessity in English (cf. Leech 2003 for British English and Kranich 2021 for American English).

More precisely, we think that our result is due to the specific licensing conditions of English *have-to*: *Have-to* needs strong evidence available in order to be acceptable. This idea is developed by Goodhue & Hacquard (2021) (see also Williams et al. in prog.), who state that *have-to* is stronger than *must*. They support this claim by contrasts as in (23), where *have-to* is infelicitous, whereas *must* is natural.

- (23) [Context: Ann has never met Jen’s husband, but sees her with a man at a party, and says to her friend Kate:]
 That *must*/?*has to* be her husband. (Goodhue & Hacquard 2021)

Williams et al. furthermore discuss the context in (24), originally introduced by Rullmann & Matthewson (2018). Rullmann & Matthewson argue, based on the contrast between (24a)/(24b), that a present TP is impossible for *had-to*. Williams et al. point out that (24c) (where *have-to* uncontroversially has a present TP), is infelicitous in this context as well. This can be explained if epistemic *have-to* is in general too strong for the context in (24).

- (24) [Context: A mother is wondering what her son got up to at a party last night. He emerges from his room holding his head and looking green. She says:]
 a. You **must have been** drunk last night.
 b. #You **had to be** drunk last night. (Rullmann & Matthewson 2018:300)
 c. #You **have to have been drunk** last night. (Williams et al., in prog.)

In our experiment, we did not specify the strength of the evidence that the detective bases their claim on; for some participants, the sentences might have therefore been judged as infelicitous based on the evidence being too weak to support the use of *have-to*. This may also be the reason why, when we compare our study to van Dooren’s (2020a), she finds much higher rate of acceptance for *had-to* in contexts forcing present TP. In her experiment, the contexts specify that there is evidence, and that it is strong.

6.2 Open questions and future directions

To our knowledge, our study is the first that experimentally assesses the acceptability of past epistemic modal sentences, a topic debated since the seventies, using parallel methods in three languages. Our design allows us to study the interaction between tense, epistemic modality, and narrative discourse, comparing the predictions of different syntactic approaches. Where do we go from here?

A first path is to investigate other modals. In this experiment, we focused on necessity modals that can inflect for tense, leaving aside overt aspectual marking. But many of the key examples discussed in the literature on modal temporal interactions involve possibility modals like *might*:

- (25) [Context: Sophie is looking for some ice cream, and checks the freezer. There is none in there. Asked why she opened the freezer, she replies:]
 There might have been ice cream in the freezer. (von Fintel & Gillies 2007:84)

Sophie seems to be making a claim about a past epistemic state she was in, which von Fintel & Gillies (2007), a.o., have used to argue in favor of the claim that epistemic modals can be interpreted in the scope of tense. Others have provided analyses in which there is an implicit *because*-clause (Hacquard 2011) or attitude (Falaus & Laca 2020) that embeds the modal. Similar to the debate surrounding FID and embedded modals in Section 2.1., under these alternative analyses, the temporal perspective is not past (relative to utterance time), but present (relative to the tense in the embedding clause). We believe studying such sentences experimentally is a promising avenue for research, especially given the cross-linguistic complexities that arise when aspect is added to the picture.

Finally, a prediction that can be tested by expanding the number of languages, is whether the lexical status of modals matters for their scopal interaction with tense. In this paper, we included two languages in which modals are (semi-)auxiliaries, and one in which modals are verbs. By including languages like German and Norwegian, for which it has been argued that at least some modals are verbs (Abraham 1999 for German; Eide 2005 for Norwegian), van Dooren's (2020b) theory can be tested in full.

Abbreviations

CLIT = clitic, DISC = discourse particle, INF = infinitive, IPFV = imperfective, PST = past, PTCP = participle

Data availability

Data, scripts used for data analysis and supplementary files are available in the following OSF repository: <https://osf.io/dya7h/>

Supplementary files

Appendix A: Experimental material: List of test and filler sentences

Appendix B: Experimental material: Instructions

Appendix C: Experimental results: Details

Appendix D: Experimental results: Betting task, full results

Appendix E: Experiment 2 ('online' judgments), response times analyses

Appendix F: AJT, comparison between Experiment 1 and 2, per language

Appendix G: Model R output from statistical analyses

Ethics and consent

The study reported as Experiment 1 was approved by IRB 326587-12. The study reported as Experiment 2 was approved by the Neurobiology of Language Department, Max Planck Institute for Psycholinguistics.

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Competing interests

The authors have no competing interests to declare.

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